

Evidence Report

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Title: Workstation Image Examination II – TSK

Case: 26-T005

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Document Revision History

Name	Revision Date	Version	Description
Tytus Gonzalez	2026-04-08	0.1	Draft
Tytus Gonzalez	2026-04-15	1.0	Final

Executive Summary

On 4/3/2026, a corporate workstation was reported to have had sensitive documents stolen and posted on a competitor's website. A forensic investigation was conducted using a forensic image of the workstation to determine the partition structure, verify the integrity of the disk, and identify relevant user and file activity associated with the incident. I, Tytus Gonzalez, prepared a Windows forensic virtual machine and conducted analysis using both WinHex and The Sleuth Kit in a controlled environment.

The investigation began with verifying the integrity of the disk image using hashing tools to ensure no corruption occurred during acquisition. The image was then opened in WinHex to perform a low-level analysis of the disk, beginning at offset 0x000. The first 440 bytes were examined to confirm the presence of boot code, and the disk signature located at offset 0x01B8 was recorded. The partition table at offset 0x01BE was analyzed and found to contain four entries, including unused entries. Partition 1 was identified as bootable and of type NTFS. The LBA start of the partition was calculated as 63. The partition size was determined to be approximately 10.7 GB and closely matches the total size of the disk image. Examination of the partition start revealed the NTFS boot sector signature, confirming the file system type and integrity. Based on these findings, the disk image was determined to use a Master Boot Record (MBR) partitioning scheme.

Following the initial disk analysis, further examination was conducted using The Sleuth Kit to analyze the NTFS file system and user activity. The partition offset of 63 sectors was used to perform file system analysis. The volume serial number was identified, and key file system parameters, including sector size (512 bytes) and cluster size (4096 bytes), were confirmed. The Master File Table (MFT) was located at cluster 786432, providing the foundation for file system metadata analysis.

User activity was examined by identifying interactive user profiles within the "Documents and Settings" directory. Three user accounts were identified: **Administrator**, **Devon**, and **Jean**. Further examination of user directories showed that Jean's Desktop contained two files, including a Microsoft Excel document (m57biz.xls), while Devon's Desktop contained no files.

The Excel file identified on Jean's Desktop was analyzed and revealed that the file was originally authored by **Alison Smith** and last saved by **Jean**, indicating user interaction with the document. The presence of this file, along with associated metadata, provides insight into user activity on the system and may be relevant to the reported data exfiltration incident.

This investigation demonstrates a comprehensive forensic analysis process, including verification of disk integrity, examination of partition structures, and detailed file system and user activity analysis. The combined use of low-level disk analysis and file system forensics provides a clear understanding of the structure, usage, and potential relevance of data present on the workstation.

Synopsis

The Professor provided the following questions.

Review the first 446 bytes

1. Does this image have boot code in the first 440 bytes?
2. Give your observations including any plaintext in the code.
3. What is the “disk signature” starting at offset 0x01B8? Provide the value as a hexadecimal value like 0x01234567; Give the correct value with endianness in consideration

Review the partition table beginning at offset 0x01BE

1. How many partition entries are in the table? Include empty ones.
2. Which partition entry is marked bootable?
3. Partition entry 1 - What is the partition entry type? Give the hexadecimal value and the type (e.g. Linux, FAT32)
4. Partition entry 1 - What is the LBA start address? Give the hexadecimal value with endianness in consideration
5. Calculate the offset of the LBA start address. Sector size is 512 bytes or 0x200 in hexadecimal.
6. Partition entry 1 - What is the partition size? Give the hexadecimal value with endianness in consideration
7. Calculate the size of the partition in bytes. Sector size is 512 bytes or 0x200 in hexadecimal Size must be in decimal
8. What is the difference, in bytes, between the actual size of the evidence file versus the value above?

Navigate to Partition Entry One LBA Start Address Offset

1. What are the first 8 bytes at this offset? DO NOT consider endianness in this value
2. Does this value support your partition type answer above?
3. Give your observations including any plaintext in the code.
4. Based on your observations, does this disk image use an MBR or GPT partitioning scheme? Explain why.

- 1. Using 'mmls' , what is the starting sector of the NTFS/exFAT partition?**
- 2. Using 'fsstat' , what is the volume serial number?**
- 3. What is the sector size?**
- 4. What is the cluster size?**
- 5. How many sectors fit into one cluster?**
- 6. What is the first cluster of the MFT?**
- 7. Using 'fls' , what is the metadata reference (inode) for the folder "Documents and Settings"?**
- 8. Using this metadata reference and 'fls' , there are three directories representing interactive users (users that have logged into the system). What are the directory (user) names? Do not include "All Users", "Default User", "Local Service", or "Network Service".**
- 9. Using 'istat' and the metadata reference for the folder "Devon", review the \$STANDARD_INFORMATION block. What is the create date timestamp?**
- 10. Do the same for the folder "Jean". What is the create date timestamp?**
- 11. Based on this information, what user profile was created first?**
- 12. Using 'fls' , review the contents of Jean's "Desktop" folder. What are the file names of the two files?**
- 13. Using 'istat' , what is the created date/timestamp of the file WITHOUT the ".url" extension?**
- 14. Are there any files in Devon's "Desktop" folder? If so, what are the file names?**
- 15. Use 'icat' to recover the file from question 13 to disk. Run the 'file' command on the recovered file. According to the file output, who was the "author" and "last saved" by entries?**

Evidence Collected

The section outlines all physical evidence items collected

Name	Identifier (Tag)	Description
windows-image-01.dd	windows-image-01	DD image of Windows machine provided by instructor

This section provides details of the digital evidence collected

Name	windows-image-01.dd
Type:	DOS/MBR boot sector MS-MBR XP english at offset 0x12c "Invalid partition table" at offset 0x144 "Error loading operating system" at offset 0x163 "Missing operating system", disk signature 0x39bf39be; partition 1 : ID=0x7, active, start-CHS (0x0,1,1), end-CHS (0x3ff,254,63), startsector 63, 20948697 sectors
Size:	10737418240 bytes
MD5:	78a52b5bac78f4e711607707ac0e3f93
SHA1:	ba7dc57e08bb6e3393aee15c713ae04feadcd181
SHA256	4d4fc46f284630a69980340ee36d3ed486de424a9f456eedba0f7194cc7a991b

Tools Used

The following outlines the workstations and all hardware and software used for this investigation.

Workstation

Hostname	Operating System	Build	Physical Virtual	/	Built
Windows 11x64_DFIR	Windows 11 Home	2021	Virtual		02/06/2026

Software

Name	Version	Release	Purpose
WinHex/X-ways Forensics	21.6	Oct 19, 2025	A Universal hexadecimal editor, particularly helpful in the realm of computer forensics, data recovery, low-level data processing, and IT security
The Sleuth kit	4.14.0	April 14, 2025	A forensic toolset used to analyze disk images, recover files, and examine file system metadata in digital investigations.

Case Information & Observables

This case involved collecting, processing, and analyzing a Windows-based corporate workstation provided by the Client. The workstation was reported to have had sensitive corporate documents stolen and posted on a competitor's website. The following are my observations and responses to Client questions regarding the disk image, partition layout, and filesystem structure.

Observables

Review the first 446 bytes

1. Does this image have boot code in the first 440 bytes?

Yes, the first 440 bytes contain boot code. This is evident because the bytes are not all zeros and include executable instructions, as well as readable text strings commonly found in MBR code.

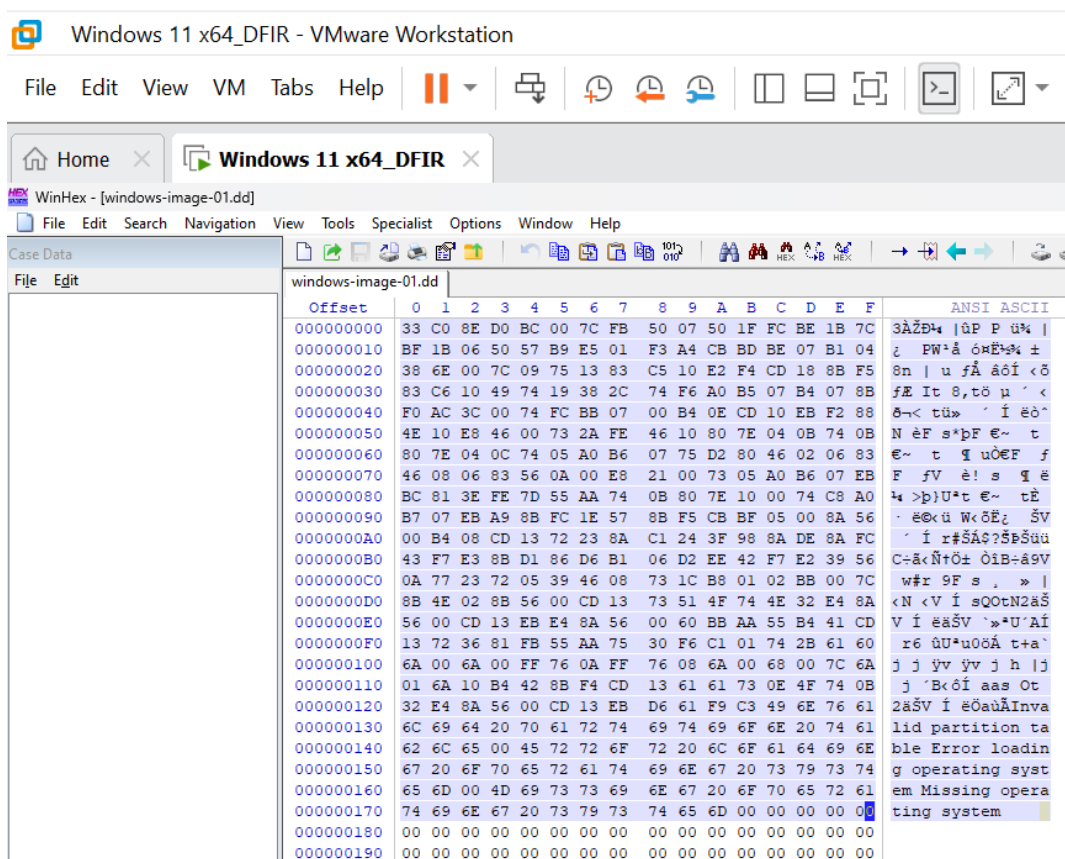


Figure 1: The hexadecimal value of the first 440 bytes of the dd; the plaintext on the side confirms that this is MBR boot code

2. Give your observations including any plaintext in the code.

Within the first 440 bytes, there are two notable plaintext strings: “Invalid partition table”, “Missing operating system”, and “Error loading operating system”. This confirms that the first 440 bytes are of MBR boot code, which is designed to display errors when the system cannot find a valid bootable partition.

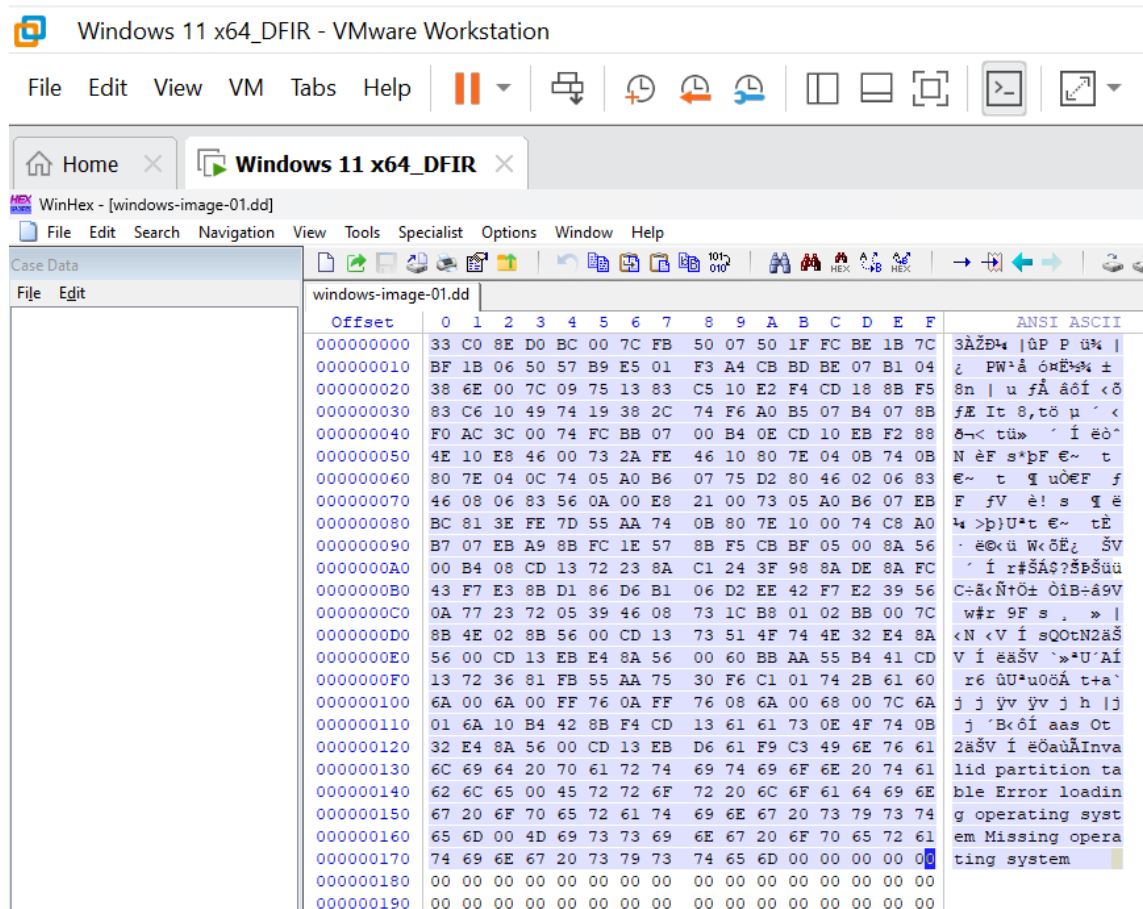


Figure 2: The hexadecimal value of the first 440 bytes of the dd; the plaintext on the side confirms that this is MBR boot code

3. What is the “disk signature” starting at offset 0x01B8? Provide the value as a hexadecimal value like 0x01234567. Give the correct value with endianness in consideration.

The disk signature located at offset 0x01B8 is 0x39BF39BE. Because the data is stored in Little-Endian format, you must read the four bytes at offset 0x01B8 and reverse their order to get the correct hexadecimal representation. This 4-byte identifier is used by the OS to distinguish between physical disks attached to the system.

0000001A0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
0000001B0	00 00 00 00 00 00 2C 44 63	BE 39 BF 39 00 00 80 01	,Dc%9ç9 €
0000001C0	01 00 07 FE FF FF 3F 00	00 00 D9 A6 3F 01 00 00	pÿÿ? Û!?
0000001D0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
0000001E0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
0000001F0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 55 AA	U*

Figure 3: The hexadecimal value of the disk signature starting at offset 0x01B8

Review the partition table beginning at offset 0x01BE

1. How many partition entries are in the table? Include empty ones.

The partition table contains 4 partition entries. This is standard for an MBR, which always allocates space for four entries regardless of whether they are all in use.

0000001A0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
0000001B0	00 00 00 00 00 00 2C 44 63	BE 39 BF 39 00 00 80 01	,Dc%9ç9 €
0000001C0	01 00 07 FE FF FF 3F 00	00 00 D9 A6 3F 01 00 00	pÿÿ? Û!?
0000001D0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
0000001E0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
0000001F0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 55 AA	U*

Figure 4: The MBR partition table at offset 0x01BE showing all 4 partition entries

2. Which partition entry is marked bootable?

Partition entry 1 is marked as bootable. This is indicated by the boot value at offset 0x01BE of 0x80 in the first byte of the entry. The partition is active and designated to be used for booting the system, if the value were 0x00 then it would be not bootable.

0000001A0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
0000001B0	00 00 00 00 00 00 2C 44 63	BE 39 BF 39 00 00 80 01	,Dc%9ç9 €
0000001C0	01 00 07 FE FF FF 3F 00	00 00 D9 A6 3F 01 00 00	pÿÿ? Û!?
0000001D0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	

Figure 5: The hexadecimal value at offset 0x01BE showing the value as 80

3. Partition entry 1 - What is the partition entry type? Give the hexadecimal value and the type (e.g. Linux, FAT32)

This partition entry is of type NTFS/exFAT. I found this given the hexadecimal value 0x07 from the value at offset 0x01BE + 4.

000000190	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0000001A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0000001B0	00	00	00	00	00	2C	44	63	BE	39	BF	39	00	00	80	01			
0000001C0	01	00	07	FE	FF	FF	3F	00	00	00	D9	A6	3F	01	00	00			

Figure 6: The MBR partition entry 1 showing the value to be 0x07

4. Partition entry 1 - What is the LBA start address? Give the hexadecimal value with endianness in consideration

The LBA start address is located at bytes 8-11 of the partition entry (offset 0x01C6 to 0x01C9). The endianness of the bytes is equivalent to 0x0000003F (which is 63 in decimal).

000000170	74	69	6E	67	20	73	79	73	74	65	6D	00	00	00	00	00	00	00	00
000000180	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000000190	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0000001A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0000001B0	00	00	00	00	00	2C	44	63	BE	39	BF	39	00	00	80	01			
0000001C0	01	00	07	FE	FF	FF	3F	00	00	00	D9	A6	3F	01	00	00			
0000001D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0000001E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

Figure 7: The hexadecimal value at offset 0x01BE + 8 which is the LBA start address

5. Calculate the offset of the LBA start address. Sector size is 512 bytes or 0x200 in hexadecimal.

The LBA start for Partition 1 is 0x0000003F which in decimal form is 63. To find the sector size, I multiply 63 * 512 = 32,256 (in hexadecimal this is 0x07E00).

6. Partition entry 1 - What is the partition size? Give the hexadecimal value with endianness in consideration

The value of sectors for partition 1 is 0x013FA6D9, which equals 20,948,697 sectors after applying little-endian conversion.

000000190	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0000001A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0000001B0	00	00	00	00	00	2C	44	63	BE	39	BF	39	00	00	80	01			
0000001C0	01	00	07	FE	FF	FF	3F	00	00	00	D9	A6	3F	01	00	00			
0000001D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0000001E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

Figure 8: The hexadecimal value of Partition 1 at offset 0x01CA-0x01CD

7. Calculate the size of the partition in bytes. Sector size is 512 bytes or 0x200 in hexadecimal Size must be in decimal

The total size of Partition 1 is 10,725,732,864 bytes, calculated as the number of sectors multiplied by 512 bytes per sector.

8. What is the difference, in bytes, between the actual size of the evidence file versus the value above?

The difference between the total disk size and the calculated partition size is 11,685,376 bytes, calculated by the number of bytes in the dd image minus the size of the partition in bytes. This difference represents space occupied by the MBR, unused sectors, and any padding within the disk image.

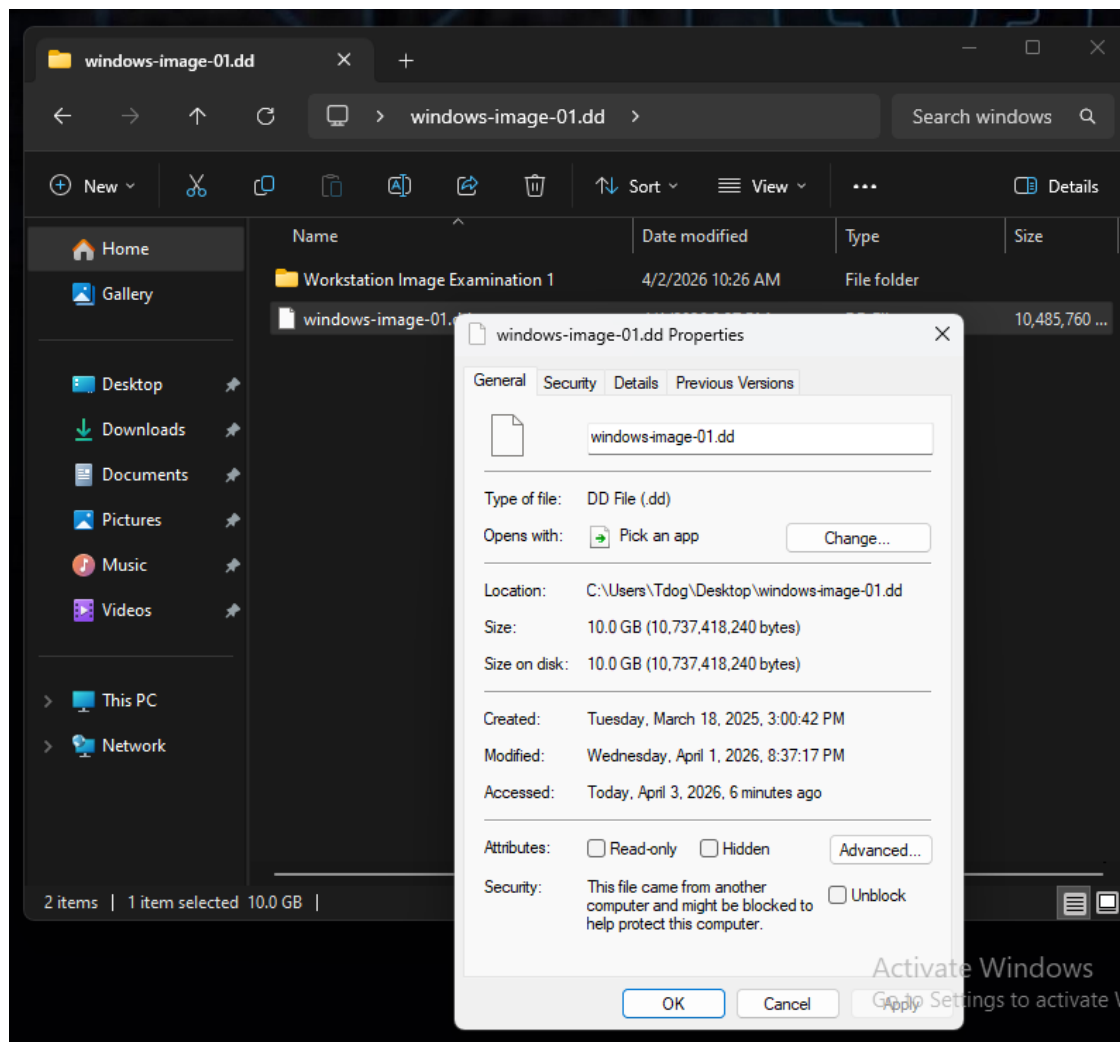


Figure 9: Screenshot of the size of the DD file

Navigate to Partition Entry One LBA Start Address Offset

1. What are the first 8 bytes at this offset? DO NOT consider endianness in this value

The first 8 bytes at the partition's starting offset (0x01BE) are: 80 01 01 00 07 FE FF FF.

2. Does this value support your partition type answer above?

Yes, because the byte at offset 0x01C2 is 07 which is the standardized Hex code for the NTFS/exFAT file system. This is a commonly used file system in Windows environments.

3. Give your observations including any plaintext in the code.

The hex dump contains standard MBR partition table fields, including a boot indicator, LBA start/end address, and the partition type. To the right of the hex code, you can see portions of an operating system string: "Missing operating system". This is a common error message found in MBR bootloader code areas.

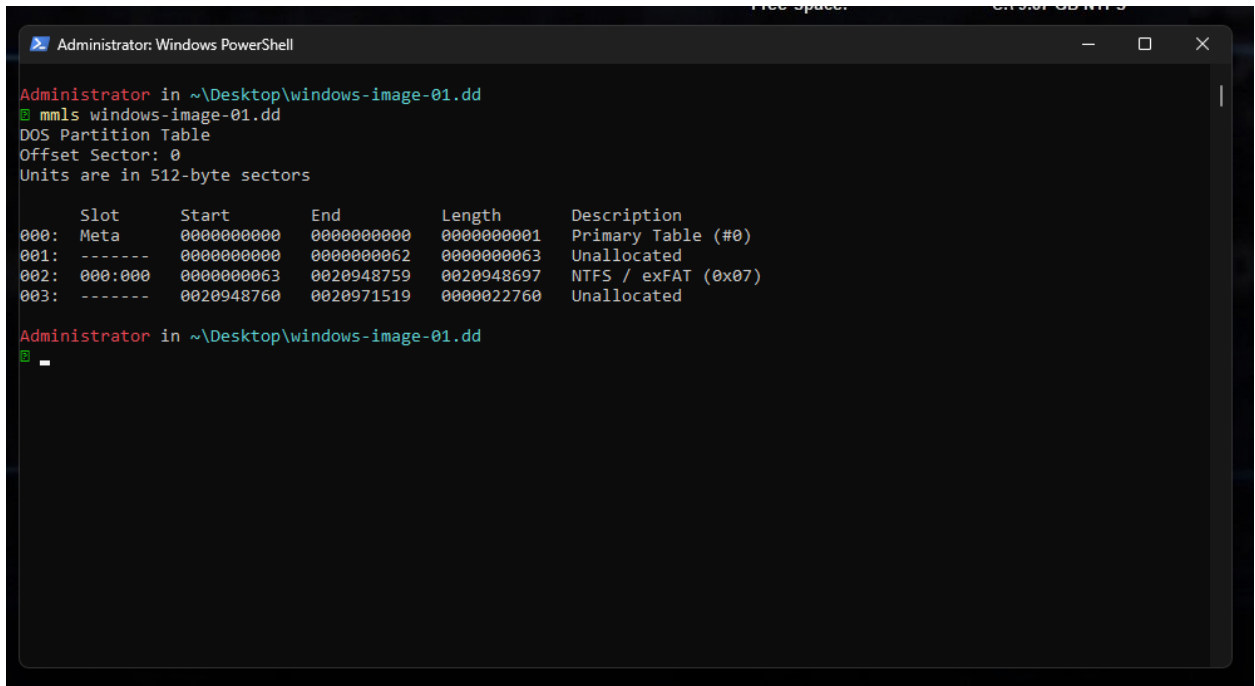
4. Based on your observations, does this disk image use an MBR or GPT partitioning scheme? Explain why.

This disk image uses an MBR partitioning scheme. The data is structured as 16-byte partition entries starting at 0x01BE. This is the exact layout of the legacy MBR partition table. A GPT partition entry is 128 bytes in size.

Examination 2 Questions

1. Using 'mmls', what is the starting sector of the NTFS/exFAT partition?

The starting sector of the NTFS/exFAT partition is 63, as identified using the 'mmls' command. This value matches the LBA start address determined during the previous WinHex analysis, where the starting sector was calculated as 0x0000003F, which is equivalent to 63 in decimal. This starting sector tells us where our partition should start for when we do further analysis.



```
Administrator: Windows PowerShell

Administrator in ~\Desktop\windows-image-01.dd
mmls windows-image-01.dd
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

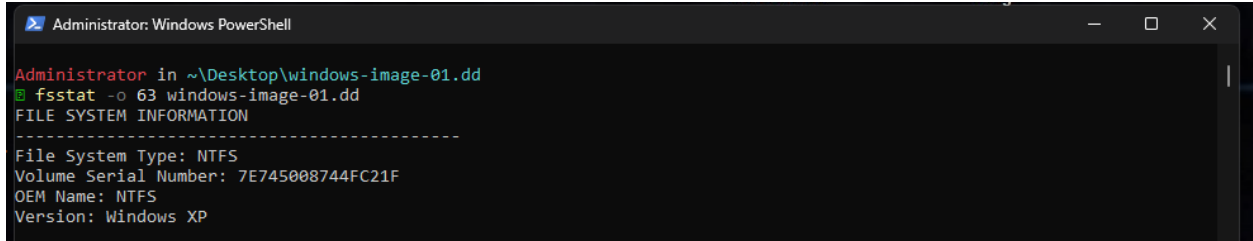
   Slot   Start      End          Length    Description
000:  Meta   0000000000  0000000000  0000000001  Primary Table (#0)
001:  ----- 0000000000  0000000062  0000000063  Unallocated
002:  000:000 0000000063  0020948759  0020948697  NTFS / exFAT (0x07)
003:  ----- 0020948760  0020971519  0000022760  Unallocated

Administrator in ~\Desktop\windows-image-01.dd
_
```

Figure 10: The Sleuth Kit 'mmls' command to find starting sector

2. Using 'fsstat', what is the volume serial number?

Using 'fsstat' I was able to find file system information and confirm that the file system was NTFS, which we already knew from the previous examination. The volume serial number was 7E745008744FC21F and this version was "Windows XP".



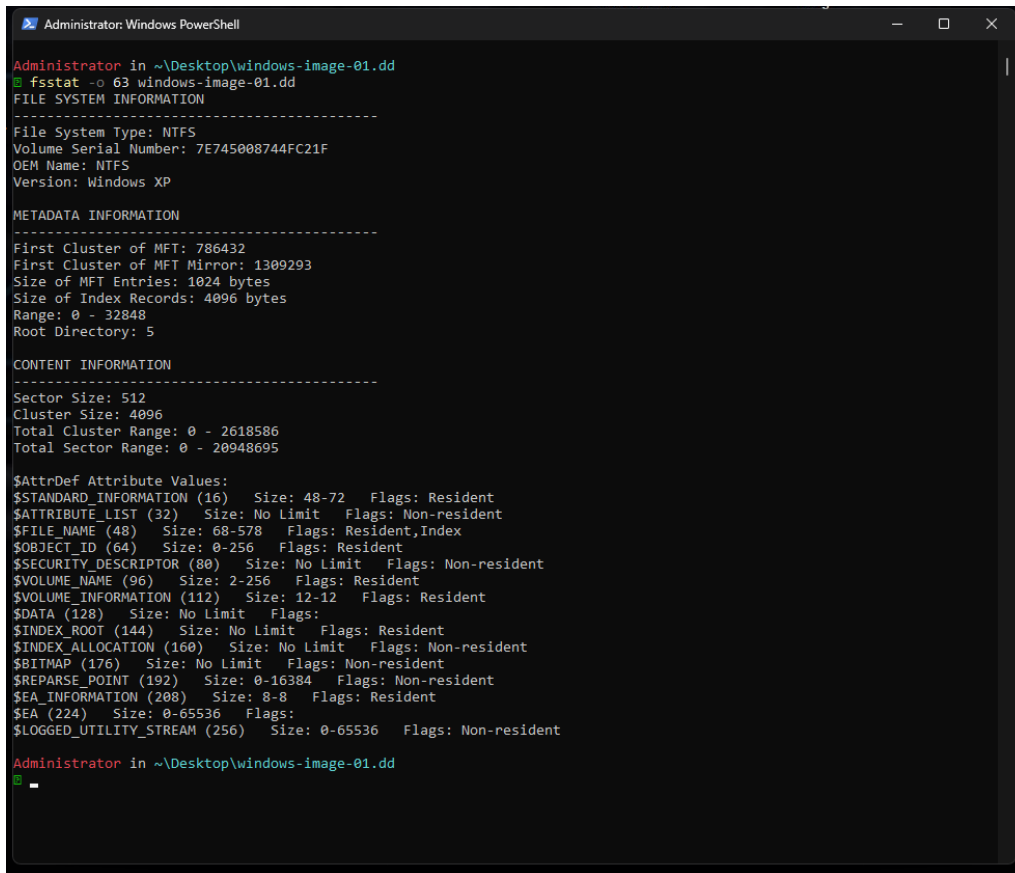
```
Administrator: Windows PowerShell

Administrator in ~\Desktop\windows-image-01.dd
fsstat -o 63 windows-image-01.dd
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: 7E745008744FC21F
OEM Name: NTFS
Version: Windows XP
```

Figure 11: The Sleuth Kit 'fsstat' command to find volume serial number

3. What is the sector size?

Using the 'fsstat' command, I was able to find the sector size to be 512 bytes. A sector is one tiny block on a disk, it is the smallest addressable unit on the physical disk used by a filesystem.



```
Administrator: Windows PowerShell

Administrator in ~\Desktop\windows-image-01.dd
fsstat -o 63 windows-image-01.dd
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: 7E745008744FC21F
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 786432
First Cluster of MFT Mirror: 1309293
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 32848
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 2618586
Total Sector Range: 0 - 20948695

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$REPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 8-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident

Administrator in ~\Desktop\windows-image-01.dd
-
```

Figure 12: The Sleuth Kit 'fsstat' command to find sector size and cluster size

4. What is the cluster size?

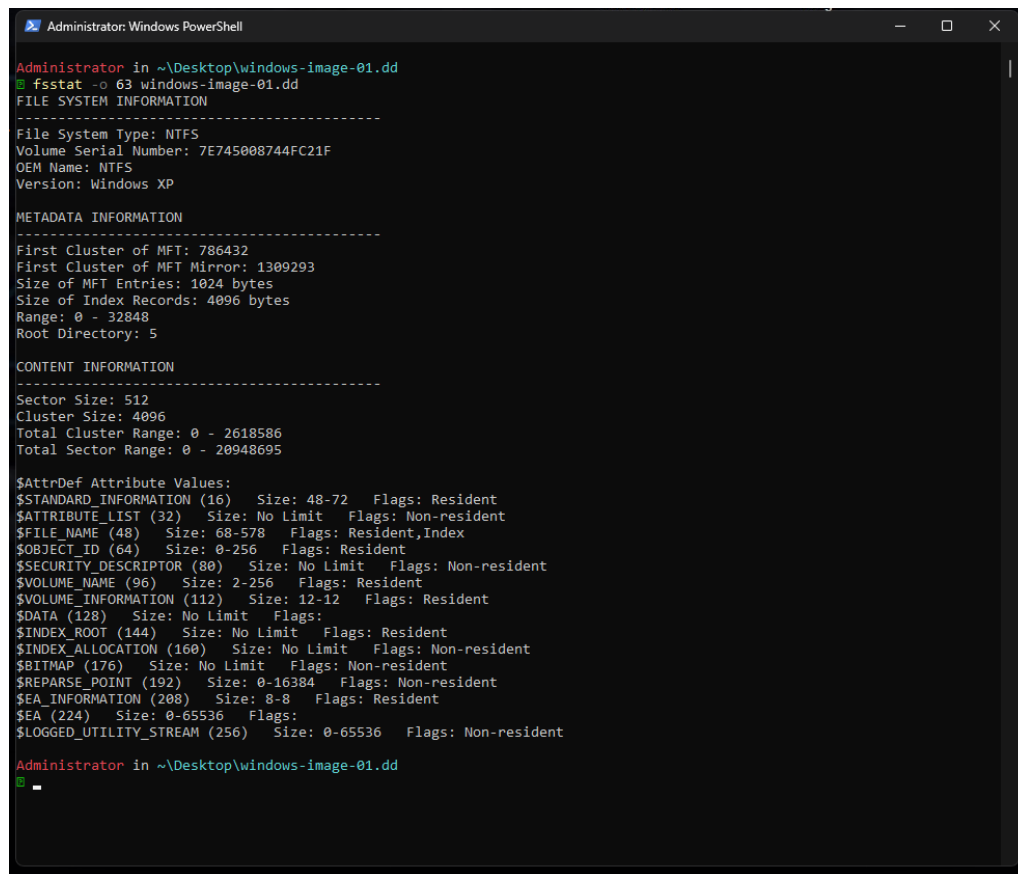
Using the 'fsstat' command, I was able to find the cluster size to be 4096 bytes. This matches the sector size of 512 bytes because 8 sectors for a cluster ($512 * 8 = 4096$), which is consistent with NTFS allocation behavior.

5. How many sectors fit into one cluster?

There are 8 sectors that belong to 1 cluster. A cluster consists of multiple sectors, allowing the filesystem to manage storage more efficiently. Therefore, if the sector size is 512 bytes, the cluster size should be $512 * 8 = 4096$ bytes.

6. What is the first cluster of the MFT?

Using the 'fsstat' command, I was able to find the MFT first cluster to be 786432. The Master File Table (MFT) is a critical structure in NTFS that stores metadata for every file and directory on the system.



```
Administrator: Windows PowerShell

Administrator in ~\Desktop\windows-image-01.dd
fsstat -o 63 windows-image-01.dd
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: 7E74508744FC21F
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 786432
First Cluster of MFT Mirror: 1309293
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 32848
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 2618586
Total Sector Range: 0 - 20948695

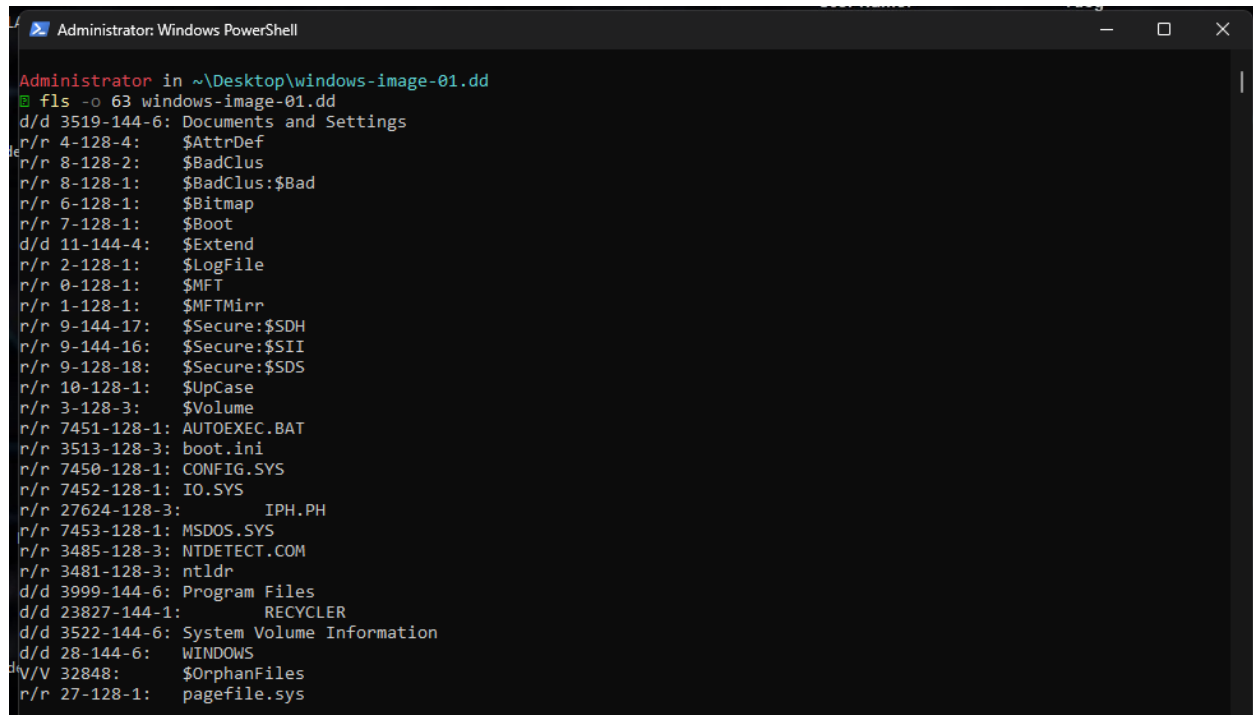
$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$REPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 8-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident

Administrator in ~\Desktop\windows-image-01.dd
```

Figure 13: The Sleuth Kit 'fsstat' command to find metadata information

7. Using 'fls', what is the metadata reference (inode) for the folder "Documents and Settings"?

The metadata reference for the folder "Documents and Settings" is inode 3519 and is a directory, as identified using the 'fls' command.



```
Administrator: Windows PowerShell
Administrator in ~\Desktop\windows-image-01.dd
fls -o 63 windows-image-01.dd
d/d 3519-144-6: Documents and Settings
r/r 4-128-4: $AttrDef
r/r 8-128-2: $BadClus
r/r 8-128-1: $BadClus:$Bad
r/r 6-128-1: $Bitmap
r/r 7-128-1: $Boot
d/d 11-144-4: $Extend
r/r 2-128-1: $LogFile
r/r 0-128-1: $MFT
r/r 1-128-1: $MFTMirr
r/r 9-144-17: $Secure:$SDH
r/r 9-144-16: $Secure:$SII
r/r 9-128-18: $Secure:$SDS
r/r 10-128-1: $UpCase
r/r 3-128-3: $Volume
r/r 7451-128-1: AUTOEXEC.BAT
r/r 3513-128-3: boot.ini
r/r 7450-128-1: CONFIG.SYS
r/r 7452-128-1: IO.SYS
r/r 27624-128-3: IPH.PH
r/r 7453-128-1: MSDOS.SYS
r/r 3485-128-3: NTDETECT.COM
r/r 3481-128-3: ntldr
d/d 3999-144-6: Program Files
d/d 23827-144-1: RECYCLER
d/d 3522-144-6: System Volume Information
d/d 28-144-6: WINDOWS
V/V 32848: $OrphanFiles
r/r 27-128-1: pagefile.sys
```

Figure 14: The Sleuth Kit 'fls' command to find the "Documents and Settings" folder

8. Using this metadata reference and 'fls', there are three directories representing interactive users (users that have logged into the system). What are the directory (user) names? Do not include "All Users", "Default User", "Local Service", or "Network Service".

Using the metadata reference for the "Documents and Settings" directory (inode 3519), the 'fls' command was used to enumerate its contents. The interactive user directories identified were **Administrator (inode 10222)**, **Devon (inode 17437)**, and **Jean (inode 16144)**. All of the users are identified as directories.

```
Administrator: Windows PowerShell
+++ r/r 22193-128-3: hand2.gif

Administrator in ~\Desktop\windows-image-01.dd
[+] fls -o 63 windows-image-01.dd
d/d 3519-144-6: Documents and Settings
r/r 4-128-4: $AttrDef
r/r 8-128-2: $BadClus
r/r 8-128-1: $BadClus:$Bad
r/r 6-128-1: $Bitmap
r/r 7-128-1: $Boot
d/d 11-144-4: $Extend
r/r 2-128-1: $LogFile
r/r 0-128-1: $MFT
r/r 1-128-1: $MFTMirr
r/r 9-144-17: $Secure:$SDH
r/r 9-144-16: $Secure:$SII
r/r 9-128-18: $Secure:$SDS
r/r 10-128-1: $UpCase
r/r 3-128-3: $Volume
r/r 7451-128-1: AUTOEXEC.BAT
r/r 3513-128-3: boot.ini
r/r 7450-128-1: CONFIG.SYS
r/r 7452-128-1: IO.SYS
r/r 27624-128-3: IPH.PH
r/r 7453-128-1: MSDOS.SYS
r/r 3485-128-3: NTDETECT.COM
r/r 3481-128-3: ntldr
d/d 3999-144-6: Program Files
d/d 23827-144-1: RECYCLER
d/d 3522-144-6: System Volume Information
d/d 28-144-6: WINDOWS
V/V 32848: $OrphanFiles
r/r 27-128-1: pagefile.sys

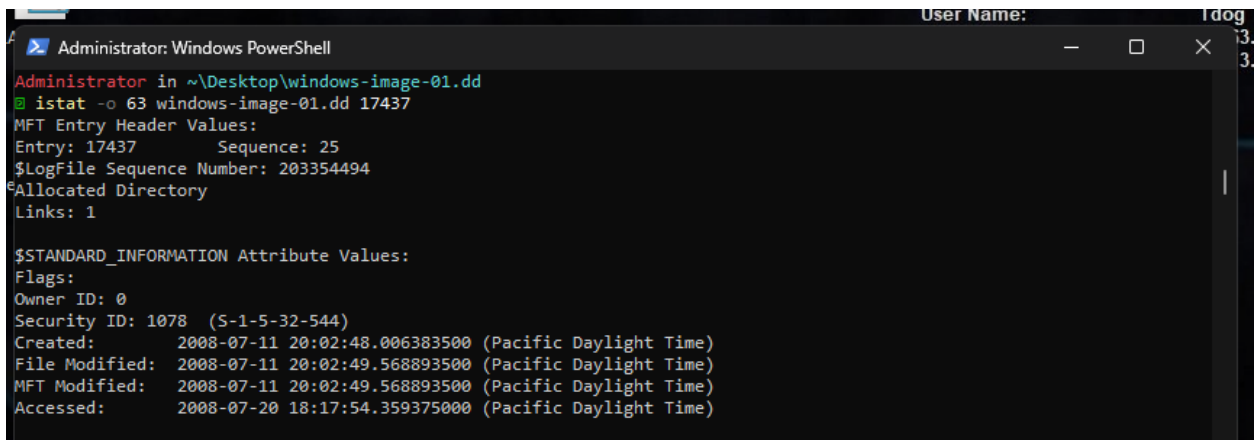
Administrator in ~\Desktop\windows-image-01.dd
[+] fls -o 63 windows-image-01.dd 3519
d/d 10222-144-6: Administrator
d/d 3521-144-6: All Users
d/d 3520-144-7: Default User
d/d 17437-144-5: Devon
d/d 16144-144-5: Jean
d/d 10151-144-6: LocalService
d/d 3368-144-6: NetworkService

Administrator in ~\Desktop\windows-image-01.dd
[+]
```

Figure 15: The Sleuth Kit 'fls' command from the "Documents and Settings" directory to find users

9. Using 'istat' and the metadata reference for the folder "Devon", review the \$STANDARD_INFORMATION block. What is the create date timestamp?

Using the 'istat' command on the metadata reference for the folder "Devon", the created timestamp from the "\$STANDARD_INFORMATION" attribute is 2008-07-11 20:02:48.006383500 (Pacific Daylight Time).



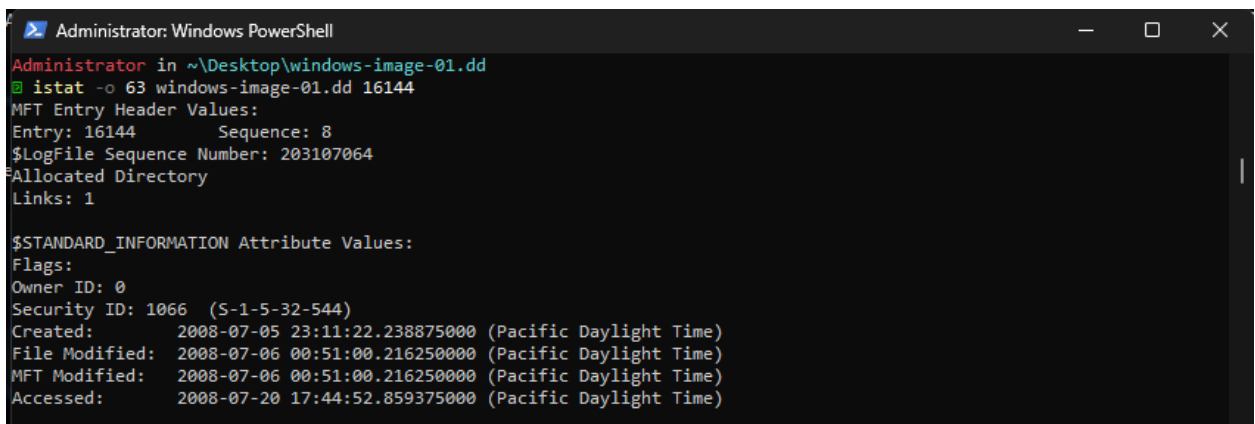
```
Administrator: Windows PowerShell
Administrator in ~\Desktop\windows-image-01.dd
istat -o 63 windows-image-01.dd 17437
MFT Entry Header Values:
Entry: 17437      Sequence: 25
$LogFile Sequence Number: 203354494
Allocated Directory
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags:
Owner ID: 0
Security ID: 1078 (S-1-5-32-544)
Created:      2008-07-11 20:02:48.006383500 (Pacific Daylight Time)
File Modified: 2008-07-11 20:02:49.568893500 (Pacific Daylight Time)
MFT Modified:  2008-07-11 20:02:49.568893500 (Pacific Daylight Time)
Accessed:      2008-07-20 18:17:54.359375000 (Pacific Daylight Time)
```

Figure 16: The Sleuth Kit 'istat' command used to obtain information about the user 'Devon'

10. Do the same for the folder "Jean". What is the create date timestamp?

Using the 'istat' command on the metadata reference for the folder "Jean", the created timestamp from the "\$STANDARD_INFORMATION" attribute is 2008-07-05 23:11:22.238875000 (Pacific Daylight Time).



```
Administrator: Windows PowerShell
Administrator in ~\Desktop\windows-image-01.dd
istat -o 63 windows-image-01.dd 16144
MFT Entry Header Values:
Entry: 16144      Sequence: 8
$LogFile Sequence Number: 203107064
Allocated Directory
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags:
Owner ID: 0
Security ID: 1066 (S-1-5-32-544)
Created:      2008-07-05 23:11:22.238875000 (Pacific Daylight Time)
File Modified: 2008-07-06 00:51:00.216250000 (Pacific Daylight Time)
MFT Modified:  2008-07-06 00:51:00.216250000 (Pacific Daylight Time)
Accessed:      2008-07-20 17:44:52.859375000 (Pacific Daylight Time)
```

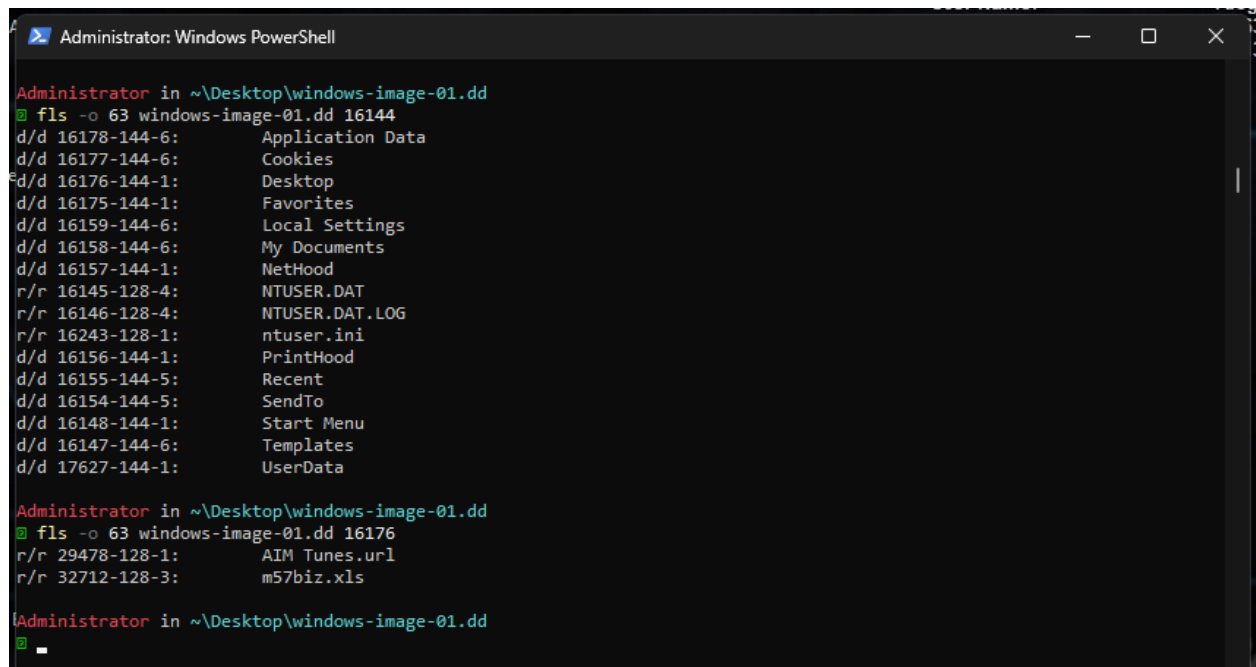
Figure 17: The Sleuth Kit 'istat' command used to obtain information about the user 'Jean'

11. Based on this information, what user profile was created first?

Based on the created timestamps from the users, "Jean" was created 6 days before "Devon".

12. Using 'fls' , review the contents of Jean's "Desktop" folder. What are the file names of the two files?

Using the 'fls' command on the metadata reference for the "Jean" user directory, the contents of the Desktop folder were examined. The two files present in the Desktop directory are 'AIM Tunes.url' and 'm57biz.xls'.



```
Administrator in ~\Desktop\windows-image-01.dd
fls -o 63 windows-image-01.dd 16144
d/d 16178-144-6:      Application Data
d/d 16177-144-6:      Cookies
d/d 16176-144-1:      Desktop
d/d 16175-144-1:      Favorites
d/d 16159-144-6:      Local Settings
d/d 16158-144-6:      My Documents
d/d 16157-144-1:      NetHood
r/r 16145-128-4:      NTUSER.DAT
r/r 16146-128-4:      NTUSER.DAT.LOG
r/r 16243-128-1:      ntuser.ini
d/d 16156-144-1:      PrintHood
d/d 16155-144-5:      Recent
d/d 16154-144-5:      SendTo
d/d 16148-144-1:      Start Menu
d/d 16147-144-6:      Templates
d/d 17627-144-1:      UserData

Administrator in ~\Desktop\windows-image-01.dd
fls -o 63 windows-image-01.dd 16176
r/r 29478-128-1:      AIM Tunes.url
r/r 32712-128-3:      m57biz.xls

Administrator in ~\Desktop\windows-image-01.dd
_
```

Figure 18: The Sleuth Kit 'fls' command used to view the contents of the Desktop directory

13. Using 'istat' , what is the created date/timestamp of the file WITHOUT the ".url" extension?

Using the 'istat' command on the file without the ".url" extension, the created timestamp from the \$STANDARD_INFORMATION attribute is 2008-07-19 18:28:03.656250000.

```
Administrator: Windows PowerShell
Type: $INDEX_ROOT (144-1) Name: $I30 Resident size: 376

Administrator in ~\Desktop\windows-image-01.dd
istat -o 63 windows-image-01.dd 32712
MFT Entry Header Values:
Entry: 32712 Sequence: 2
$LogFile Sequence Number: 172434688
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 1067 (S-1-5-21-484763869-796845957-839522115-1004)
Created: 2008-07-19 18:28:03.656250000 (Pacific Daylight Time)
File Modified: 2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified: 2008-07-19 18:28:04.046875000 (Pacific Daylight Time)
Accessed: 2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: m57biz.xls
Parent MFT Entry: 16176 Sequence: 3
Allocated Size: 294912 Actual Size: 291840
Created: 2008-07-19 18:28:03.687500000 (Pacific Daylight Time)
File Modified: 2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified: 2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
Accessed: 2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$OBJECT_ID Attribute Values:
Object Id: d9cb701e-55fa-11dd-ad9a-000c298d352b

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-4) Name: N/A Resident size: 86
Type: $OBJECT_ID (64-5) Name: N/A Resident size: 16
Type: $DATA (128-3) Name: N/A Non-Resident size: 291840 init_size: 291840
645029 650567 650568 522189 522190 522191 522192 522193
509952 509953 509954 509955 502852 502853 502854 502855
502856 502857 502858 502859 502860 502861 502862 502863
606339 606340 606341 606342 606343 606344 606345 606346
606347 606348 606349 606350 606351 606352 606353 606354
606355 606356 606357 606358 606359 606360 606361 606362
138035 138036 138037 138038 138039 138040 138041 138042
138043 138044 138045 138046 138047 138048 138049 138050
138051 652833 1288030 1288031 1288032 1288033 1376229 585037

Administrator in ~\Desktop\windows-image-01.dd
```

Figure 19: The Sleuth Kit 'istat' command used to obtain information about the m57biz.xls file

14. Are there any files in Devon's "Desktop" folder? If so, what are the file names?

Using the 'fls' command to examine the contents of Devon's "Desktop" directory (inode 23955), no files or directories were identified. Therefore, the Desktop folder for the user Devon does not contain any files.

```
Administrator: Windows PowerShell
d/d 3368-144-6: NetworkService

Administrator in ~\Desktop\windows-image-01.dd
fls -o 63 windows-image-01.dd 17437
d/d 23957-144-1: Application Data
d/d 23956-144-1: Cookies
d/d 23955-144-1: Desktop
d/d 23954-144-1: Favorites
d/d 23936-144-6: Local Settings
d/d 23935-144-6: My Documents
d/d 23934-144-1: NetHood
r/r 23922-128-4: NTUSER.DAT
r/r 23923-128-4: NTUSER.DAT.LOG
r/r 24050-128-1: ntuser.ini
d/d 23933-144-1: PrintHood
d/d 23932-144-1: Recent
d/d 23931-144-5: SendTo
d/d 23925-144-1: Start Menu
d/d 23924-144-6: Templates

Administrator in ~\Desktop\windows-image-01.dd
fls -o 63 windows-image-01.dd 23955

Administrator in ~\Desktop\windows-image-01.dd
```

Figure 20: The Sleuth Kit 'fls' command used to find the contents of the Desktop directory for "Devon"

15. Use 'icat' to recover the file from question 13 to disk. Run the 'file' command on the recovered file. According to the file output, who was the "author" and "last saved" by entries?

Using the 'icat' command, the file identified in Question 13 was successfully recovered from the disk image and written to disk. The recovered file was analyzed using string extraction to identify embedded metadata. Based on the output, the 'author' of the file is **Alison Smith**, and the 'last saved by' entry is **Jean**.

```

d`b
Alison Smith
lis
lis
Jean User
Microsoft Excel
`%Z%
" WMFC
EMF
j%v
0`v
j%#
j%#
0dv
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
WMFC
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
Verdana
"System
M57.biz company
Name
Position
Salary
Alison
Smithn
Presidento
$140,000
Jean
Jones0
CFO
$120,000
R%P%R%
R%P%R%
M57.BIZ
Sheet1
Worksheets
_PID_LINKBASE
j%K

Administrator in ~\Desktop\windows-image-01.dd took 4s

```

Figure 21: Using the 'strings' command to

Conclusion

This assignment provided valuable experience in performing both low-level disk analysis and file system analysis using WinHex and The Sleuth Kit. It improved my understanding of partition structures, NTFS metadata, and how to identify user activity through forensic tools. One challenge was transitioning between manual hexadecimal analysis and automated tools, but this strengthened my ability to interpret forensic data accurately. Overall, this assignment demonstrated a realistic forensic workflow and reinforced the importance of attention to detail and clear documentation.

References

“WinHex: Features and Ways of Application.” *X-Ways.net*, x-ways.net/winhex/allfeatures.html.

“Help Documents.” *The Sleuth Kit Wiki*, 2026, wiki.sleuthkit.org/Help-Documents/. Accessed 15 Apr. 2026.